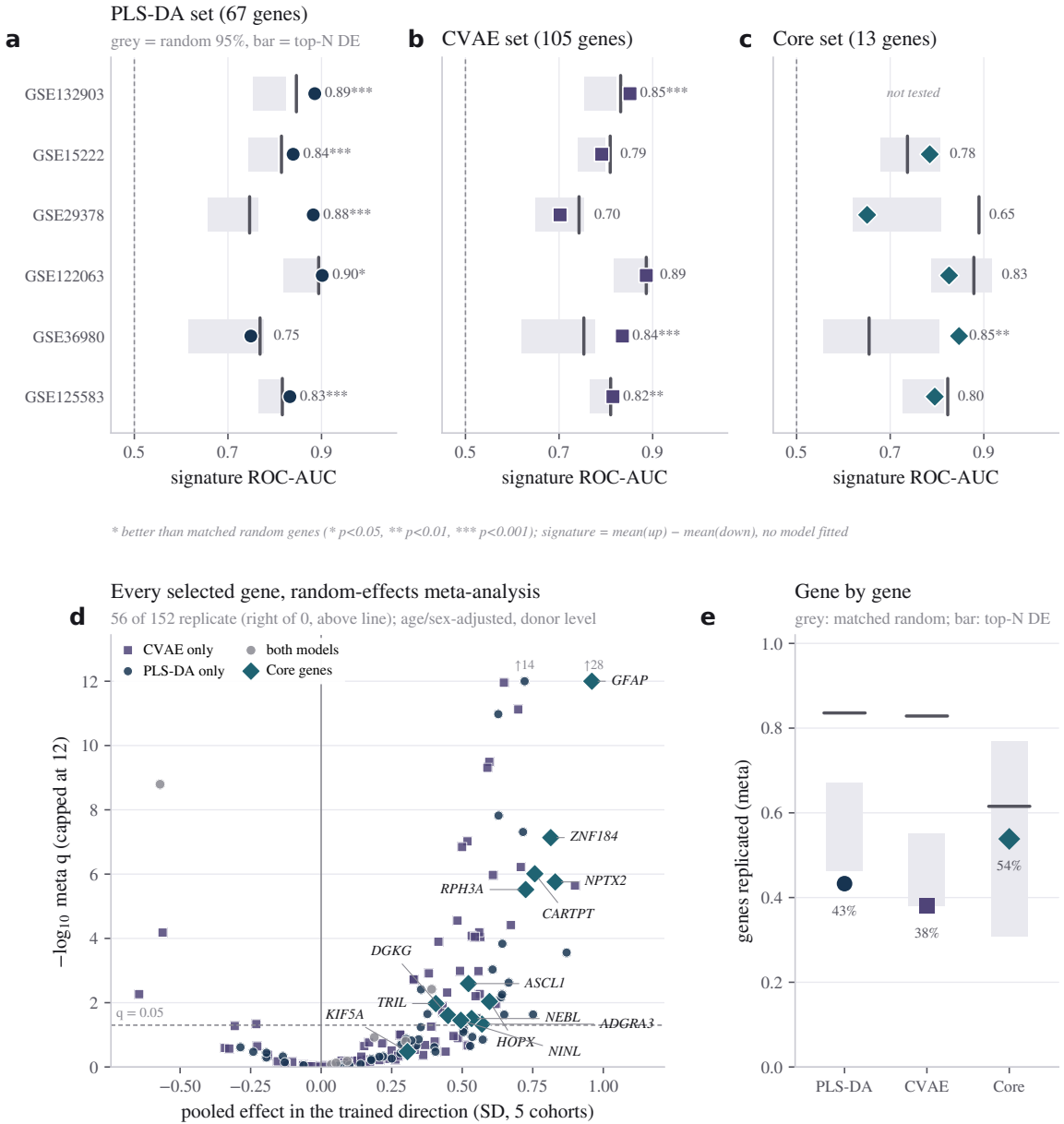
a
PLS-DA set (67 genes)
grey = random 95%, bar = top-N DE
GSE132903
GSE15222
GSE29378
GSE122063
GSE36980
GSE125583
0.89***
0.84***
0.88***
0.90*
0.75
0.83***
0.5
0.7
0.9
signature ROC-AUC
b
CVAE set (105 genes)
0.85***
0.79
0.70
0.89
0.84***
0.82**
c
Core set (13 genes)
not tested
0.78
0.65
0.83
0.85**
0.80
* better than matched random genes (* p<0.05, ** p<0.01, *** p<0.001); signature = mean(up) − mean(down), no model fitted
d
Every selected gene, random-effects meta-analysis
56 of 152 replicate (right of 0, above line); age/sex-adjusted, donor level
CVAE only
PLS-DA only
both models
Core genes
↑14
↑28
GFAP
ZNF184
NPTX2
RPH3A
CARTPT
DGKG
ASCL1
TRIL
KIF5A
NEBL
ADGRA3
HOPX
NINL
q = 0.05
−log10 meta q (capped at 12)
pooled effect in the trained direction (SD, 5 cohorts)
e
Gene by gene
grey: matched random; bar: top-N DE
genes replicated (meta)
43%
38%
54%
PLS-DA
CVAE
Core